

Supplemental Material

Title: Small Left Ventricle is Associated with an Increased Cardiovascular Risk in Patients with Atrial Fibrillation

Contents	Page number
Echocardiographic parameters measurement	2
Prosperity score matching	2
Supplemental Table 1. Definition of outcomes	3
Supplemental Table 2. Definition of comorbidities	4
Supplemental Table 3. Comparison of auxiliary echocardiographic parameters among groups with different LV size	6
Supplemental Table 4. Comparison of heart failure, inflammation biomarkers and blood component biomarkers among groups with different LV size	7
Supplemental Table 5. The antithrombotic type and dose among groups with different LV size	8
Supplemental Table 6. Proportions of causes of death among groups with different LV size	9
Supplemental Table 7. The incidence of major bleeding stratified by the anticoagulation type and LV size	10
Supplemental Table 8. The association between LV size and cardiovascular outcomes using Chinese Han cutoff values	11
Supplemental Table 9. Baseline characteristics of prosperity score matched cohort	12
Supplemental Table 10. The association between LV size and cardiovascular outcomes in the prosperity score matched cohort	14
Supplemental Table 11. The association between LV size and cardiovascular outcomes in competitive risk regression analysis	15
Supplemental Figure 1. Directed acyclic graph	16
Supplemental Figure 2 The prevalence of heart failure subtypes stratified by LV size	17
Supplemental Figure 3. Adjusted association of BSA-indexed LVEDD and cardiovascular outcomes	18
Supplemental Figure 4. Adjusted association of LVEDD and cardiovascular outcomes excluding patients with CAD	19

Supplemental Methods

Echocardiographic parameters measurement

The measurement of echocardiographic parameters was in accordance the American Society of Echocardiography (ASE) guidelines. The assessment and documentation of key echocardiographic variables was determined by the ACC/AHA AF clinical variable standards. The left ventricular diameters were measured at end ventricular diastole (LVEDD) or systole (LVESD) by two dimensional (2D) or M-mode echocardiography. The measurement of left atrial diameter used the “leading edge to leading edge” method, in centimeters, measured from M-mode in the parasternal long-axis view at end-ventricular systole. The LVEF was measured using the biplane method of disks (modified Simpson’s rule) approach. The left ventricular end-diastolic thickness of septal and posterior walls as measured in the parasternal long-axis view.

Prosperity score matching

In order to attenuate confounding of patient characteristics and exposure to LV size (interested exposure), the prosperity score matching (PSM) was performed to further validate the main results. The PS is estimated using a logistic regression model that incorporates all variables that may be related to the outcome and/or the exposure, regardless of their statistical significance or collinearity.^[1] Therefore, more variables were included in the model for calculating PS (sensitivity analysis) than those in the traditional covariates adjusted models (main analysis). The variables in the PS calculation model included age, sex, body mass index, systolic blood pressure, diastolic blood pressure, atrial fibrillation type, heart failure, previous ischemic stroke, hypertension, coronary artery disease, diabetes, chronic kidney disease, hyperlipidemia, peripheral arterial disease, catheter ablation, smoking status, drinking status, left atrial diameter, interventricular septum thickness, left ventricular posterior wall thickness, oral anticoagulant use, antiplatelet drugs use, anti-arrhythmic drugs use, angiotensin-converting enzyme inhibitors /angiotensin receptor blockers use, and statin use. Then, the 1-to-2 PSM was performed in which a patient in the small LV group was matched with a patient in the normal LV group by means of the nearest-neighbor method without replacement, with a caliber set at 0.1. For the matched cohort, the baseline characteristics were compared between small LV and matched normal LV groups to evaluate the covariates balance. The mixed effect Cox proportional model was performed to compare the survival free from outcomes between small LV and matched normal LV subgroups, with the pair as a mixed effect. Further adjustment for covariates in the Cox model was waived due to the limited event incidence and reduced statistical power.

References :

1. Comparison of Propensity Score Methods and Covariate Adjustment: Evaluation in 4 Cardiovascular Studies. *J Am Coll Cardiol*. 2017 Jan 24;69(3):345-357. doi: 10.1016/j.jacc.2016.10.060.

Supplemental Table 1. Definition of outcomes

Outcomes	Definition
Cardiovascular death	Cardiovascular (CV) death includes death resulting from an acute myocardial infarction (MI), sudden cardiac death, death due to heart failure (HF), death due to stroke, death due to CV hemorrhage, and death due to other CV causes. Further specifications include: • Myocardial infarction • Ischemic stroke • Heart failure • Intracranial hemorrhage • Non–intracranial hemorrhage-related death • Unexplained sudden death • Other cardiovascular causes
Ischemic stroke	Documented stroke or cerebrovascular accident consisting of acute loss of neurological function caused by an ischemic event with residual symptoms at least 24 hours after onset.
Systematic embolism	Abrupt vascular insufficiency associated with clinical and radiological evidence of arterial occlusion in a vascular bed other than the cerebrovascular system in the absence of other likely mechanisms (e.g., atherosclerosis). • limb artery • visceral-mesenteric artery • renal artery • splenic arteries • other peripheral arteries
Major bleeding	Bleeding leads to: • Transfusion of at least 2 units of whole blood or erythrocytes • Requiring hospitalization or surgery • Resulting in permanent disability • Involving a critical anatomic site (retroperitoneal, pericardial, intra-spinal, intracranial, non-traumatic intra-articular, or intra-ocular bleeding associated with abrupt deterioration of visual acuity)

Supplemental Table 2. Definition of comorbidities

Comorbidities	Definition
Heart failure	• Physician documentation or report of any of the following symptoms of heart failure before this care encounter, described as dyspnea, fluid retention, or low cardiac output secondary to cardiac dysfunction; or the

	description of rales, jugular venous distension, or pulmonary edema, defined by the Framingham Heart Study • A previous hospital admission with principal diagnosis of heart failure • Echocardiographic assessed left ventricular ejection fraction $\leq 40\%$
Ischemic stroke	Documented history of stroke or cerebrovascular accident with acute loss of neurological function caused by an ischemic or hemorrhagic event with residual symptoms at least 24 hours after onset. List most likely etiology: • Larger-artery disease (e.g., carotid) • Small-artery disease (lacunar) • Embolism • Other, specify • Not specified
Hypertension	• History of hypertension diagnosed and treated with medication, diet, and/or exercise • Blood pressure greater than or equal to 140 mmHg systolic or 90 mmHg diastolic on at least 2 occasions • Currently undergoing antihypertensive pharmacological therapy.
Coronary artery disease	• Previous myocardial infarction (hospital admission, electrocardiogram report, consistent increase in biochemical marker, patient reports history) • Prior coronary artery bypass surgery • Prior percutaneous coronary intervention • Angiographically documented coronary artery stenosis greater than or equal to 50% • Positive stress test; specify imaging modality if performed • Angina pectoris
Diabetes	History of diabetes, regardless of duration of disease, need for antidiabetic agents, or a fasting blood sugar greater than or equal to 7 mmol/l or 126 mg/dl
Chronic kidney disease	The presence of kidney damage or an estimated glomerular filtration rate less than 60 ml/min/1.73 m², persisting for 3 months or more
Hyperlipidemia	Previously diagnosed with dyslipidemia and/or treated by a physician
Peripheral vascular disease	Previous history of peripheral vascular disease (includes upper and lower extremity, renal, mesenteric, and abdominal aortic systems)

Supplemental Table 3. Comparison of auxiliary echocardiographic parameters among groups with different LV size

	Normal LV	Small LV	Large LV	<i>P</i> value
RVd, mm, N	21.0 (19.0,23.0), 3465	20.0 (19.0,23.0), 180*	22.0 (20.0,24.0), 158	< 0.001
RVOTd, mm, N	27.0 (25.0,30.0), 3564	26.0 (24.0,28.0), 187*	29.0 (26.0,31.0), 164	< 0.001
TRVmax, N	3704	197	171	
Not detected	1264 (34.1)	65 (33.0) *	35 (20.5)	< 0.001
≤280cm/s	2143 (57.9)	121 (61.4) *	111 (64.9)	
280~340cm/s	276 (7.5)	11 (5.6) *	21 (12.3)	
>340cm/s	21 (0.6)	0 (0.0) *	4 (2.3)	
SPAP, N	3704	197	171	
Not detected	1196 (32.3)	65 (33.0) *	38 (22.2)	0.004
≤35mmHg	2008 (54.2)	110 (55.8) *	95 (55.6)	
>35mmHg	500 (13.5)	22 (11.2) *	38 (22.2)	
MPAd, mm, N	23.0 (21.0,24.0), 3618	21.0 (20.0,23.0), 189*	24.0 (22.0,26.0), 167	< 0.001
RAAd, mm, N	42.0 (39.0,45.0), 1998	40.0 (37.0,42.0), 84*	44.0 (40.5,48.0), 135	< 0.001
PA peak velocity, cm/s, N	88.0 (78.0,100.0), 3602	86.5 (75.2,96.0), 190	86.0 (71.0,99.0), 165	0.019
Aortic peak velocity, cm/s, N	120.0 (105.0,135.0), 3564	114.0 (100.0,131.0), 187 *	120.0 (103.0,138.8), 164	0.007
E wave, cm/s, N	84.0 (70.0,100.0), 3656	84.0 (65.8,102.2), 192	92.0 (78.0,107.5), 167	0.001
A wave, cm/s, N	75.0 (60.0,90.0), 2004	75.0 (56.0,87.2), 96	74.0 (59.0,83.0), 61	0.364
E/A, N	1.0 (0.7,1.4), 1996	0.9 (0.7,1.4), 95	1.1 (0.8,1.4), 60	0.151

* There is statistically significant difference ($P<0.05$) between normal LV and small groups.

LV = left ventricle; RVd = right ventricle diameter, RVOTd = right ventricle outflow tract diameter; TRVmax = maximal tricuspid regurgitation jet velocity; SPAP = systolic pulmonary artery pressure; MPAd = main pulmonary artery diameter; RAAd = right atria diameter; PA = pulmonary artery.

Supplemental Table 4. Comparison of heart failure, inflammation biomarkers and blood component biomarkers among groups with different LV size

	Normal LV (N=7027)	Small LV (N=308)	Large LV (N=429)	<i>P</i> value
BNP, pg/mL, N	113.0 (49.0,239.0), 4185	112.0 (57.0,271.5), 181	243.0 (125.0,591.0), 228	< 0.001
hsCRP, mg/L, N	0.9 (0.4,2.1), 4929	0.9 (0.4,2.1), 219	1.2 (0.5,3.4), 201	< 0.001
Hb, g/L, N	145.8 (16.3), 4995	148.3 (16.9) *, 221	144.7 (18.0), 221	0.044
RBC, 10 ¹² /L, N	4.7 (0.5), 4995	4.8 (0.5) *, 221	4.7 (0.6), 221	0.017
WBC, 10 ⁹ /L, N	6.2 (5.2,7.4), 4995	6.5 (5.5,7.5), 221	6.4 (5.4,7.7), 221	0.050
PLT, 10 ⁹ /L, N	204.5 (53.0), 4995	201.7 (53.2), 221	207.5 (58.2), 221	0.514

* There is statistically significant difference ($P<0.05$) between normal LV and small groups.

Values are presented as mean (standard deviation) or median (IQR).

LV = left ventricle; BNP = B-type natriuretic peptide; hsCRP = high-sensitivity C reactive protein; Hb = hemoglobin; RBC = red blood cell; WBC = white blood cell; PLT = platelet.

Supplemental Table 5. The antithrombotic type and dose among groups with different LV size

	Normal LV (N=7027)	Small LV (N=308)	Large LV (N=429)	<i>P</i> value
Antiplatelet	953 (13.6)	51 (16.6)	115 (26.8)	< 0.001
OAC	5623 (80)	244 (79.2)	287 (66.9)	< 0.001
Warfarin	3328 (47.4)	140 (45.5)	204 (47.6)	0.802
NOAC	2295 (32.7)	104 (33.8)	83 (19.3)	< 0.001
Rivaroxaban	696 (9.9)	32 (10.4)	21 (4.9)	
20mg QD	26 (0.4)	2 (0.7)	2 (0.5)	
15mg QD	659 (9.4)	29 (9.4)	19 (4.4)	
10mg QD	11 (0.2)	1 (0.3)	0 (0)	
Dabigatran	1598 (22.7)	71 (23.1)	61 (14.2)	
150mg BID	58 (0.8)	0 (0)	2 (0.5)	
110mg BID	1540 (21.9)	71 (23.1)	59 (13.8)	
Apixaban 5mg BID	1 (0.01)	1 (0.3)	1 (0.2)	

No statistically significant difference between the small LV and normal LV groups (all $P>0.05$).

LV = left ventricle; OAC = oral anticoagulant; NOAC = novel anticoagulant.

Supplemental Table 6. Proportions of causes of death among groups with different LV size

	Normal LV (N=7027)	Small LV (N=308)	Large LV (N=429)
Total Death Number	374	25	68
Death Cause			
Non-CV cause	206(55.1)	9(36.0)	28(41.2)
Heart failure	56(15.0)	6(24.0)	16(23.5)
Sudden cardiac death	27(7.2)	2(8.0)	9(13.2)
Myocardial infarction	32(8.6)	3(12.0)	6(8.8)
Ischemic stroke	19(5.1)	2(8.0)	3(4.4)
Intracranial hemorrhage	20(5.3)	3(12.0)	5(7.4)
Other CV cause	14(3.7)	0(0.0)	1(1.5)

LV = left ventricle; CV = cardiovascular.

Supplemental Table 7. The incidence of major bleeding stratified by the anticoagulation type and LV size

		No-OAC	Warfarin	NOAC Low Dose	NOAC Standard Dose
Normal LV	MB event	18	45	16	1
	Total Population	1404	3328	2218	77
	MB Incidence, per 1000 person-years	3.31(1.96-5.23)	3.39(2.47-4.54)	1.86(1.06-3.02)	3.30(0.08-18.39)
Small LV	MB event	2	4	1	0
	Total Population	64	140	101	3
	MB Incidence, per 1000 person-years	8.51(1.03-30.74)	7.13(1.94-18.26)	2.70(0.07-15.02)	null
Large LV	MB event	4	4	0	0
	Total Population	142	204	78	5
	MB Incidence, per 1000 person-years	7.49(2.04-19.18)	5.13(1.4-13.13)	null	null

LV = left ventricle; OAC = oral anticoagulant; NOAC = novel oral anticoagulant; MB = major bleeding.

Supplemental Table 8. The association between LV size and cardiovascular outcomes using Chinese Han cutoff values※

	Events/Person-years	Incidence rate (95%CI), per 1000 person-years	Unadjusted HR (95%CI)	P value	Adjusted HR (95%CI)	P value
Primary Outcome ^a						
Normal LV	443/25331	17.49 (15.9-19.2)	Ref		Ref	
Small LV	17/335	50.82 (29.56-81.25)	2.92 (1.80-4.75)	0.005	2.65 (1.62-4.33)	<0.001
Large LV	112/4068	27.53 (22.67-33.13)	1.58 (1.28-1.94)	<0.001	1.17 (0.94-1.45)	0.173
CV death						
Normal LV	157/25910	6.06 (5.15-7.08)	Ref		Ref	
Small LV	9/357	25.24 (11.53-47.86)	4.18 (2.14-8.19)	<0.001	3.40 (1.71-6.77)	0.001
Large LV	58/4186	13.85 (10.52-17.91)	2.28 (1.69-3.09)	<0.001	1.45 (1.05-2.00)	0.024
IS/SE						
Normal LV	271/25485	10.63 (9.41-11.98)	Ref		Ref	
Small LV	7/346	20.24 (8.13-41.68)	1.91 (0.90-4.04)	0.454	1.84 (0.86-3.91)	0.114
Large LV	52/4117	12.63 (9.43-16.56)	1.19 (0.88-1.60)	0.064	0.98 (0.72-1.34)	0.900
Major Bleeding						
Normal LV	74/25873	2.86 (2.25-3.59)	Ref		Ref	
Small LV	3/358	8.38 (1.73-24.49)	2.95 (0.93-9.37)	0.067	2.80 (0.88-8.97)	0.082
Large LV	18/4192	4.29 (2.54-6.79)	1.51 (0.90-2.52)	0.153	1.29 (0.75-2.21)	0.361

※ The reference cutoff values for the Chinese Han population (LVEDD: 38.4 to 54.0 mm in men and 36.7 to 49.7 mm in women) were used to define normal LV size

^a The primary outcome was the composite of incident CV death, IS/SE, or major bleeding.

LV = left ventricle; CV death = cardiovascular death; IS =ischemic stroke; SE = systematic embolism; HR = hazard ratio; CI = confidence interval.

Supplemental Table 9. Baseline characteristics of propensity score matched cohort

	Normal LV (N=616)	Small LV (N=308)	P value
Age, years	66 (59,74)	65 (57,73)	0.716
Woman	149 (24.2)	68 (22.1)	0.476
Height, cm	170 (164,175)	170 (162,173)	0.102
Weight, kg	70 (60,76)	67.8 (60,75)	0.204
Body mass index, kg/m ²	24.2 (21.8,26.0)	24.2 (22.1,26.0)	0.686
Systolic blood pressure, mmHg	124 (120,134)	121 (117,135)	0.310
Diastolic blood pressure, mmHg	79 (70,80)	78 (70,81)	0.972
Persistent AF	178 (28.9)	96 (31.2)	0.476
Medical history			
Heart failure	48 (7.8)	26 (8.4)	0.732
Ischemic stroke	90 (14.6)	48 (15.6)	0.695
Hypertension	326 (52.9)	162 (52.6)	0.926
CAD	78 (12.7)	37 (12.0)	0.778
Diabetes	144 (23.4)	71 (23.1)	0.912
CKD	18 (2.9)	12 (3.9)	0.431
Hyperlipidemia	164 (26.6)	80 (26.0)	0.833
PAD	8 (1.3)	3 (1.0)	0.760
Echocardiography			
LA diameter, mm	38.0 (34.0,42.0)	38.0 (33.0,42.0)	0.938
LVEF, %	66 (62,70)	65 (61,70)	0.094
LVEDD, mm	47.0 (44.0,50.0)	40.0 (37.0,41.0)	< 0.001
LVESD, mm	30.0 (27.0,32.0)	26.0 (23.0,27.0)	< 0.001
IVS, mm	9.6 (9.0,10.0)	9.2 (8.0,11.0)	0.483
LVPW, mm	9.0 (8.0,10.0)	9.0 (8.0,10.0)	0.104
Current smoking	89 (14.4)	41 (13.3)	0.640
Current alcohol drinking	88 (14.3)	48 (15.6)	0.599
Catheter ablation	361 (58.6)	191 (62.0)	0.319
Medications			
OAC	473 (76.8)	244 (79.2)	0.403
Antiplatelet	111 (18.0)	51 (16.6)	0.582
AAD	282 (45.8)	148 (48.1)	0.514
β blocker	235 (38.1)	122 (39.6)	0.667
CCB	15 (2.4)	6 (1.9)	0.640
Digoxin	20 (3.2)	12 (3.9)	0.611
ACEI/ARB	140 (22.7)	71 (23.1)	0.912
Statin	207 (33.6)	110 (35.7)	0.524

Values are presented as n (%) or median (IQR).

LV = left ventricle; BMI = body mass index; SBP = systolic blood pressure; DBP = diastolic blood pressure; AF = atrial fibrillation; CAD = coronary heart disease; CKD = chronic kidney disease; PAD = peripheral artery disease; LA = left atrium; LVEF =

left ventricular ejection fraction; LVEDD = left ventricular end-diastolic dimension; LVESD = left ventricular end-systolic dimension; IVS = interventricular septum; LVPW = left ventricular posterior wall; OAC = oral anticoagulant; AAD = anti-arrhythmic drugs; ACEI = angiotensin-converting enzyme inhibitors; ARB = angiotensin receptor blockers.

Supplemental Table 10. The association between LV size and cardiovascular outcomes in the propensity score matched cohort

	Events/Person-years	Incidence rate (95%CI), per 1000 person-years	Unadjusted HR (95%CI)	<i>P</i> value
Primary Outcome ^a				
Normal LV	42/2345	17.91 (12.91-24.21)	Ref	
Small LV	33/1135	29.06 (20.01-40.83)	1.63 (1.03-2.57)	0.036
CV death				
Normal LV	12/2404	4.99 (2.58-8.72)	Ref	
Small LV	16/1180	13.55 (7.75-22.02)	2.75 (1.30-5.83)	0.008
IS/SE				
Normal LV	24/2363	10.16 (6.51-15.11)	Ref	
Small LV	15/1155	12.98 (7.27-21.42)	1.28 (0.67-2.45)	0.446
Major Bleeding				
Normal LV	8/2395	3.34 (1.44-6.58)	Ref	
Small LV	7/1177	5.95 (2.39-12.25)	1.81 (0.65-5.00)	0.255

^a The primary outcome was the composite of incident CV death, IS/SE, or major bleeding.

LV = left ventricle; CV death = cardiovascular death; IS =ischemic stroke; SE = systematic embolism; HR = hazard ratio; CI = confidence interval.

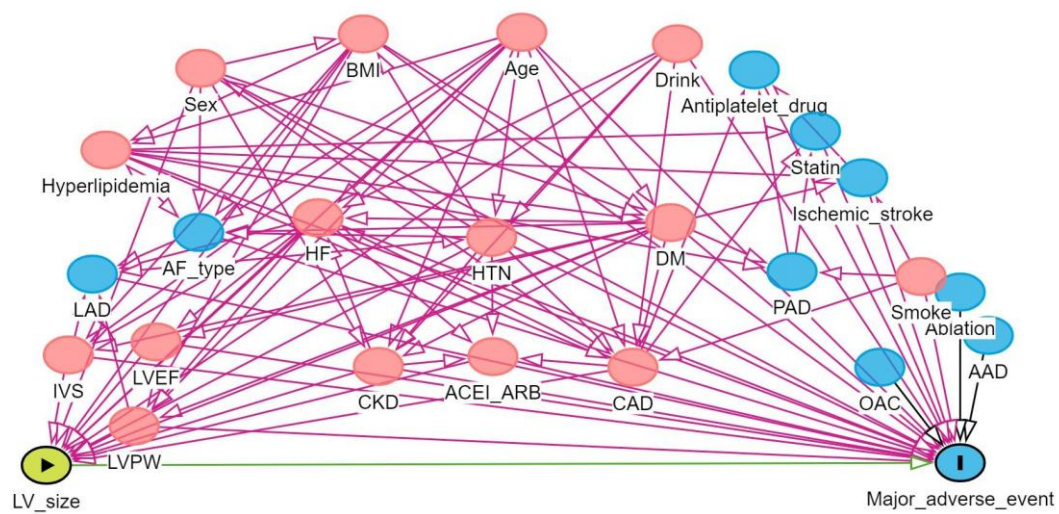
Supplemental Table 11. The association between LV size and cardiovascular outcomes in competitive risk regression analysis ^a

Outcome	Unadjusted subdistribution HR (95%CI)	<i>P</i> value	Adjusted subdistribution HR (95%CI)	<i>P</i> value
IS/SE				
Normal LV	Ref		Ref	
Small LV	1.19(0.71-2.01)	0.500	1.11(0.65-1.89)	0.710
Large LV	1.39(0.92-2.1)	0.110	1.06(0.67-1.65)	0.810
Major bleeding				
Normal LV	Ref		Ref	
Small LV	2.03(0.93-4.4)	0.074	2.13(0.97-4.64)	0.058
Large LV	1.6(0.78-3.31)	0.200	1.36(0.62-2.98)	0.440

^a The Fine-Gray model was used to take death as a competing risk.

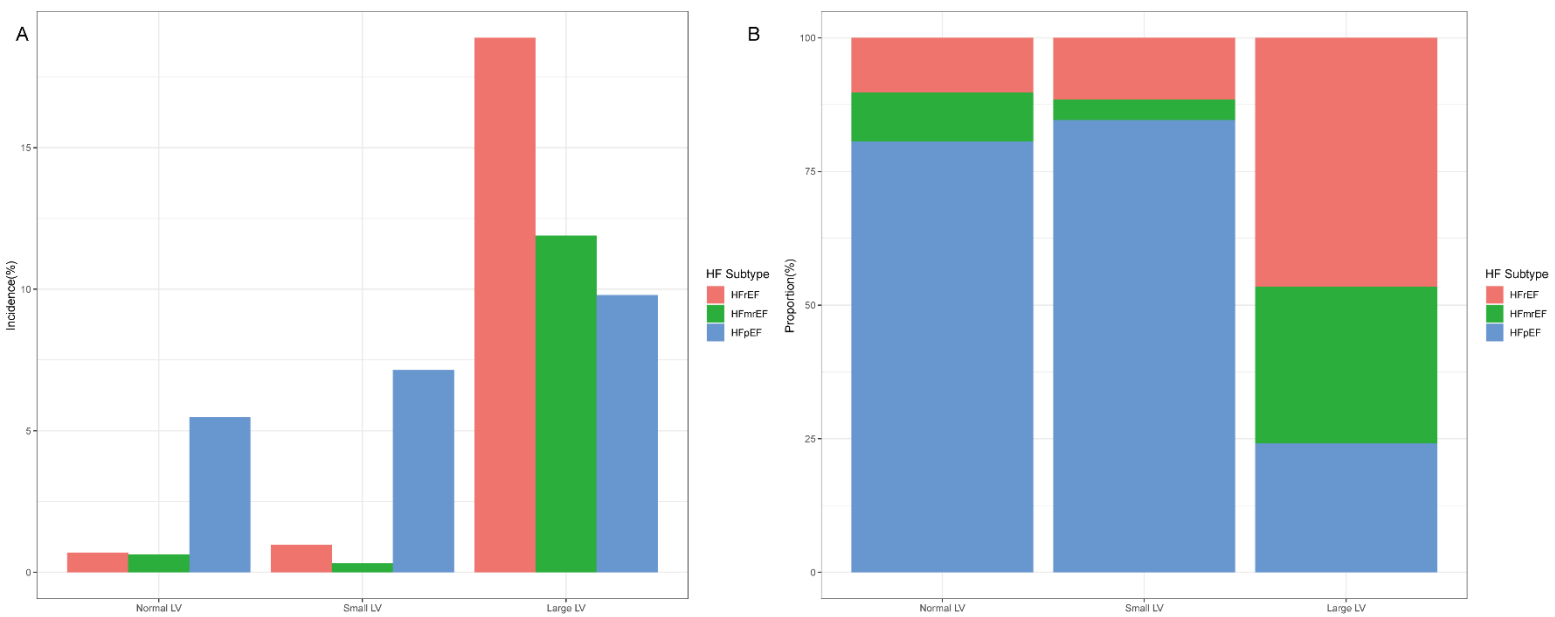
LV = left ventricle; ; IS =ischemic stroke; SE = systematic embolism; HR = hazard ratio; CI = confidence interval.

Supplemental Figure 1. Directed acyclic graph



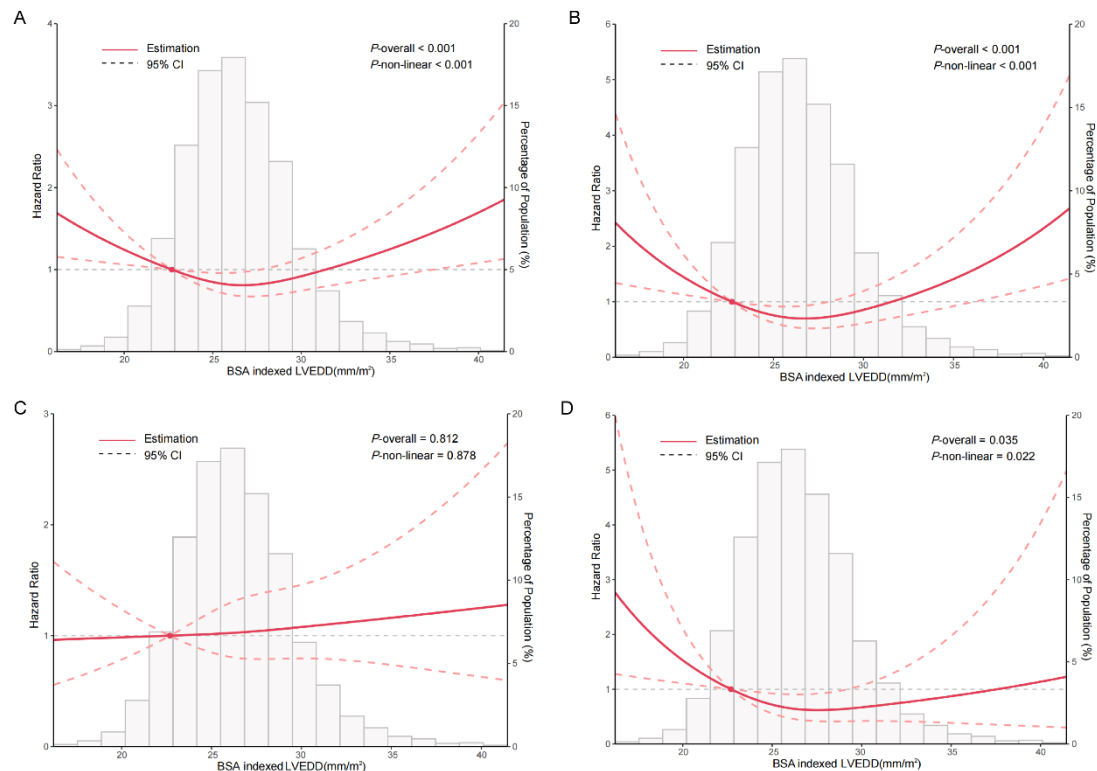
LV = left ventricle; BMI = body mass index; AF = atrial fibrillation; CAD = coronary heart disease; CKD = chronic kidney disease; DM = diabetes mellitus; PAD = peripheral artery disease; HF = heart failure; LAD = left atrial diameter; LVEF = left ventricular ejection fraction; LVEDD = left ventricular end-diastolic dimension; IVS = interventricular septum; LVPW = left ventricular posterior wall; OAC = oral anticoagulant; AAD = anti-arrhythmic drugs; ACEI = angiotensin-converting enzyme inhibitors; ARB = angiotensin receptor blockers.

Supplemental Figure 2 The prevalence of heart failure subtypes stratified by LV size



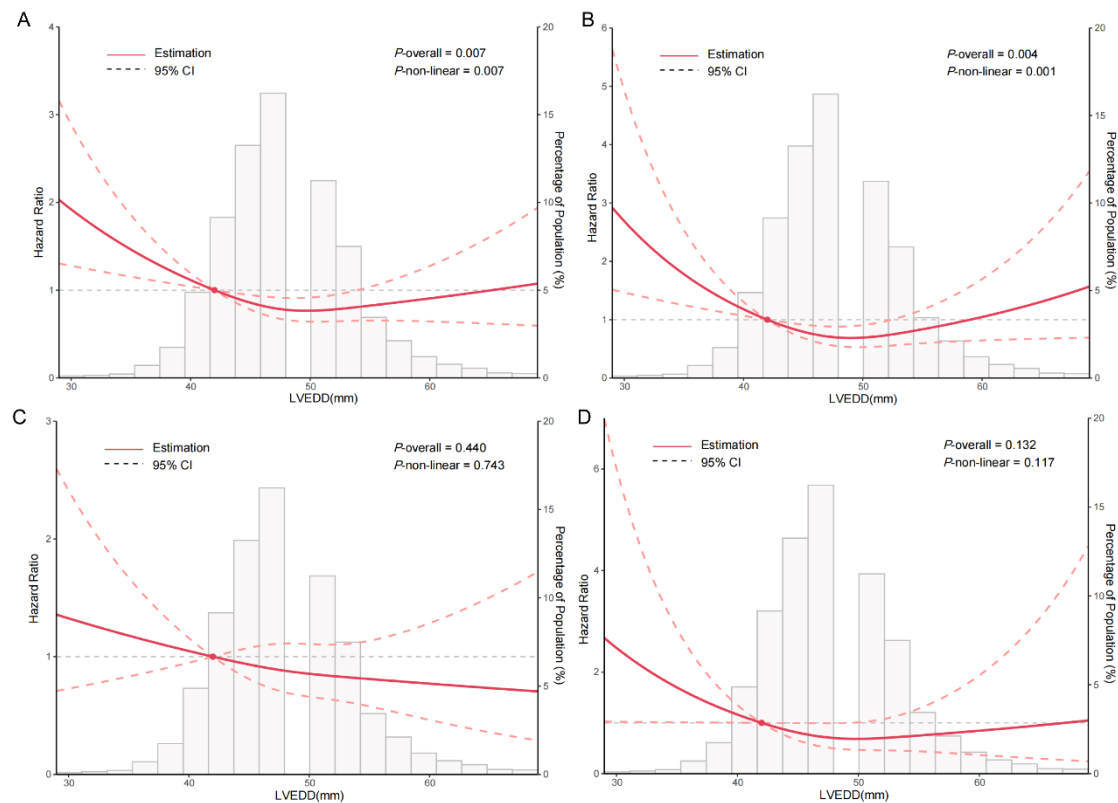
LV = left ventricle; HFrEF = heart failure with reduced ejection fraction; HFmrEF = heart failure with moderate reduced ejection fraction; HFpEF = heart failure with preserved ejection fraction.

Supplemental Figure 3. Adjusted association of BSA-indexed LVEDD and cardiovascular outcomes



Adjusted associations of measures of BSA-indexed LVEDD and incident primary outcome (composite of cardiovascular death, ischemic stroke or systematic embolism, and major bleeding) (A), CV death (B), ischemic stroke or systematic embolism (C), and major bleeding (D). The reference value (HR=1) was set at the 10th percentile of the BAS-indexed LVEDD. The solid lines represented the adjusted HRs of LVEDD across the whole range and the shaded areas represented the corresponding 95% CIs. The adjusted HRs were calculated in multivariable Cox proportional model. BSA = body surface area; LVEDD = left ventricular end-diastolic dimension; CV death = cardiovascular death; HR = hazard ratio; CI = confidence interval.

Supplemental Figure 4. Adjusted association of LVEDD and cardiovascular outcomes excluding patients with CAD



Adjusted associations of measures of LVEDD and incident primary outcome (composite of cardiovascular death, ischemic stroke or systematic embolism, and major bleeding) (A), CV death (B), ischemic stroke or systematic embolism (C), and major bleeding (D) when excluding patients with CAD. The reference value (HR=1) was set at the 10th percentile of the LVEDD. The solid lines represented the adjusted HRs of LVEDD across the whole range and the shaded areas represented the corresponding 95% CIs. The adjusted HRs were calculated in multivariable Cox proportional model. CAD = coronary heart disease; LVEDD = left ventricular end-diastolic dimension; CV death = cardiovascular death; HR = hazard ratio; CI = confidence interval.